

Facility Location Optimization for Retail Distribution in Sri Lanka Using PuLP

1 Introduction

Facility location decisions are important in logistics and supply chain management because warehouse placement directly affects transportation cost and operational efficiency. Companies must determine where to locate facilities and how to distribute products efficiently to customers.

The Facility Location Problem (FLP) is a common optimization problem used to identify the best warehouse locations while minimizing total operational costs. This project develops an Integer Linear Programming (ILP) model using Python and the PuLP library to optimize warehouse selection and store assignments in Sri Lanka. The model considers warehouse opening costs, transportation costs, warehouse capacities, and store demand levels. Geographic distance calculations and Folium visualization were also used to support the analysis.

2 Objectives

The objectives of this project are:

- Minimize warehouse and transportation costs
- Determine the optimal warehouses to open
- Assign stores efficiently to warehouses
- Satisfy warehouse capacity constraints
- Visualize the optimized network geographically

3 Problem Description

A retail company operates stores across Sri Lanka and plans to establish warehouses in selected cities. Opening warehouses creates fixed operational costs while transportation between warehouses and stores creates logistics costs. Warehouses also have limited monthly capacities. Therefore, the company must determine:

- Which warehouses should be opened
- Which stores should be assigned to each warehouse

while minimizing the total network cost.

4 Network Structure

The supply chain network consists of:

- 4 candidate warehouses
- 10 retail stores

Warehouses act as supply nodes and stores act as demand nodes.

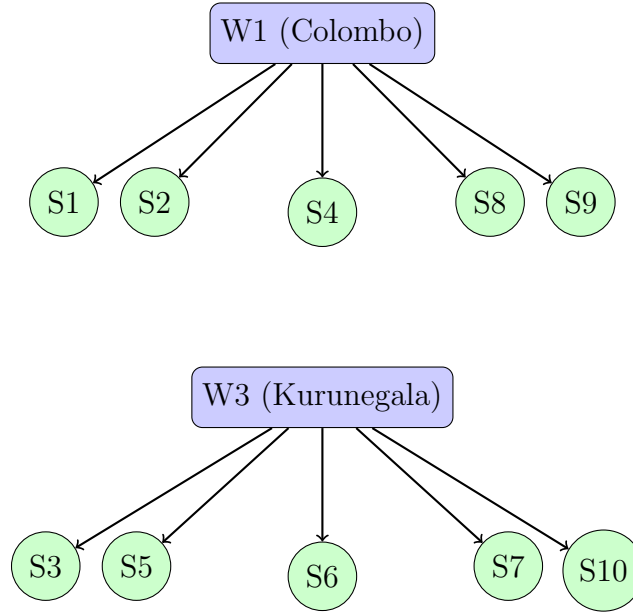


Figure 1: Optimized facility location network

The network diagram above illustrates the optimized warehouse-store assignments generated by the model.

5 Methodology

The model was implemented using Integer Linear Programming in Python.

Tool	Purpose
Python	Programming language
PuLP	Optimization modeling
Pandas	Data handling
Geopy	Distance calculations
Folium	Interactive map visualization

Table 1: Software and libraries used

Transportation costs were calculated using geographical distances between warehouses and stores. For simplicity, all monetary values are represented in LKR thousands.

6 Mathematical Formulation

6.1 Objective Function

The objective of the model is to minimize total warehouse and transportation costs.

$$\min Z = \sum_w F_w y_w + \sum_w \sum_s C_{ws} x_{ws} \quad (1)$$

Where:

- F_w = Fixed warehouse cost
- C_{ws} = Transportation cost from warehouse w to store s
- y_w = Binary variable indicating whether warehouse w is opened
- x_{ws} = Binary variable indicating whether store s is assigned to warehouse w

6.2 Constraints

1. Single Assignment Constraint

Each store must be assigned to exactly one warehouse.

$$\sum_w x_{ws} = 1 \quad (2)$$

2. Logical Assignment Constraint

Stores can only be assigned to warehouses that are opened.

$$x_{ws} \leq y_w \quad (3)$$

3. Capacity Constraint

The total assigned demand cannot exceed warehouse capacity.

$$\sum_s d_s x_{ws} \leq \text{Capacity}_w \cdot y_w \quad (4)$$

7 Data and Assumptions

Warehouse	Fixed Cost	Capacity (tons)
W1	5000	450
W2	3500	300
W3	4000	500
W4	3000	250

Table 2: Warehouse costs and capacities

Store	Monthly Demand (tons)
S1	80
S2	70
S3	120
S4	75
S5	90
S6	110
S7	85
S8	60
S9	95
S10	65

Table 3: Estimated monthly store demand

Key Assumptions

- All monetary values are represented in LKR thousands
- Warehouse capacities represent estimated monthly handling capacity
- Store demand values represent estimated monthly product demand
- Transportation cost was calculated using geographic distance
- A transportation rate of LKR 2,000 per kilometer was assumed
- Costs and demand values were estimated for modeling purposes

8 Results

The optimization model generated an optimal solution by opening two warehouses.

Warehouse	Status
W1 (Colombo)	Open
W3 (Kurunegala)	Open
W2 (Kandy)	Not Selected
W4 (Galle)	Not Selected

Table 4: Optimal warehouse selection

Store	Warehouse	Transport Cost (LKR '000)
S1	W1	62.45
S2	W1	46.78
S4	W1	78.71
S8	W1	212.85
S9	W1	328.27
S3	W3	198.89
S5	W3	106.68
S6	W3	182.73
S7	W3	151.57
S10	W3	68.08

Table 5: Store assignments and transportation costs

Minimum Total Cost: LKR 10.44 million (approximately)

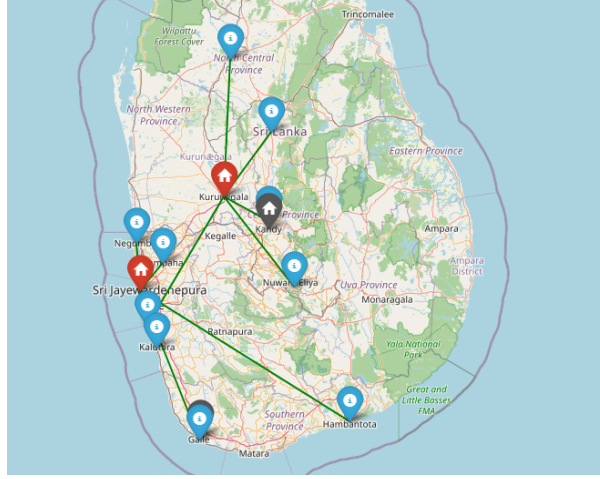


Figure 2: optimized facility location map

9 Discussion

The optimization results show that **opening only two warehouses** minimized the overall network cost while satisfying all constraints.

- Warehouse W1 (Colombo) was selected because of its strong geographic position in the western region, allowing efficient service to stores such as Negombo, Gampaha, Kalutara, Matara, and Hambantota.
- Warehouse W3 (Kurunegala) was selected because of its central location, which reduced transportation costs for stores located in the north-central and hill-country regions.
- Warehouse W2 (Kandy) and W4 (Galle) were not selected because their inclusion increased total fixed costs without producing sufficient transportation savings.

The model also demonstrated how capacity constraints influence warehouse selection and store assignments within a logistics network.

10 Conclusion

This project demonstrated how Integer Linear Programming can be applied to solve a facility location optimization problem. The model successfully minimized warehouse opening and transportation costs while satisfying assignment and capacity constraints. The optimized network selected warehouses in:

- Colombo
- Kurunegala

with a minimum total cost of approximately **LKR 10.44 million**.

References

- [1] PuLP Documentation. <https://coin-or.github.io/pulp/>
- [2] Folium Documentation. <https://python-visualization.github.io/folium/>
- [3] Geopy Documentation. <https://geopy.readthedocs.io/>
- [4] Python Software Foundation. <https://www.python.org/>